Solving Systems using Matrices



A **Matrix** is a rectangular array of numbers arranged in rows and columns.

They can be used to represent and solve linear systems.

To Convert a linear system to matrix notation:

1) Each row will correspond to an _____.

2) Each column will correspond to a _____.

A linear system of equations can be written in a matrix form as follows:

$$2x - 6y + 4z = 11$$
$$x - 3y + 4z = -7$$
$$8x + 18y - 2z = -1$$

Example 1: Write each of the following linear systems in augmented matrix form.

a) $5x - y + 3z = 2$
$x - 4y + z = -5$
$2x + 5y - 2z = -1$

b) $x + 3z - 1 = 0$
$x - 4y + 5z = 0$

Example 2: Solve the following linear system using matrices.

$$2x - 5y + z = 3$$
$$x - 3y - 2z = -9$$
$$3x - 6y - 2z = -8$$

Row-Echelon Form:

A matrix form in which each subsequent row has its first non-zero entry further to the right than in the preceding row.

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 6 & 7 & 8 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

Some of the operations include:

1. _____ two rows
2. Replace a row with a _____ _____ of itself
3. Replace a row with a _____ _____ of any two others

Examples:

$$\begin{bmatrix} 8 & 2 & 5 & -3 \\ 0 & 2 & 9 & 1 \\ 0 & 0 & -4 & 2 \\ 0 & 0 & 0 & 7 \end{bmatrix} \qquad \left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{array}\right]$$

Example 3: Solve the following linear system using matrices.

$$3x + 5y - z = 6$$
$$2x + y + 4z = 4$$
$$6x + 10y - 2z = 12$$